PROBLEM CL5

Turbojet on a Testbench: Exit State

Original statement · CL5.pdf

Moving boundaries | Mass, momentum, energy | Heat and work | Thrust

OFFICIAL PROBLEM STATEMENT

The entrance and exit sections of the turboreactor sketched in the figure of CL5.pdf have areas A_1 and A_2. Besides the outer fixed frame, the machine has mobile parts in its interior that extract an amount of work W per unit time from the fluid stream. The combined effects of chemical reaction in the combustion chamber and heat transfer through the walls give a heat transfer per unit time Q to the fluid stream. The axial force on the turboreactor measured on the testbench is $\bar{F} = -F\,\bar{e}_x$. For given values of W, Q and F and assuming that at the entrance and exit sections the flow properties are uniform, obtain the values of the density ρ_2, pressure p_2 and velocity v_2 in terms of their values p_1, ρ_1 and v_1 at the entrance. (Handwritten addition: no gravity.)

Source: CL5.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eq. 3.10, integral momentum balance Eq. 3.35, integral total energy balance Eq. 3.46, force on a body Eq. 3.32.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Compressible flow; neglect gravity; uniform properties at inlet and outlet; steady on average (the rotating parts make the flow periodic inside, but there is no net accumulation).
- Perfect gas, $h = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$, as in the class solution. Fuel mass flow neglected (not given), so the same mass flow crosses both sections.
- $\dot{W}$: work per unit time extracted from the fluid by the mobile parts; $\dot{Q}$: heat per unit time transferred to the fluid; $\bar{F} = -F\bar{e}_x$: force on the device measured on the testbench (fluid inside plus ambient air outside, at p_a).
- Data: $[\rho_1,\ p_1,\ \bar{v}_1 = v_1\bar{e}_x,\ A_1,\ A_2,\ p_a,\ F,\ \dot{W},\ \dot{Q},\ \gamma]$. Unknowns: $[\rho_2,\ p_2,\ \bar{v}_2 = v_2\bar{e}_x]$.

Symbol	Meaning	SI unit
A_1, A_2	entrance and exit areas	m^2
p_1, ρ_1, v_1	entrance state	Pa, kg/m^3, m/s
p_2, ρ_2, v_2	exit state	Pa, kg/m^3, m/s
F	thrust measured on the bench	N
$\dot{W}$, $\dot{Q}$	power extracted and heat added	W
$\dot{m} = \rho_1 v_1 A_1$	mass flow	kg/s

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the testbench; $\bar{e}_x$ along the engine axis in the flow direction.
- Surfaces: Σ_1 (entrance, $\bar{n} = -\bar{e}_x$), Σ_2 (exit, $\bar{n} = +\bar{e}_x$), Σ_w (fixed walls, $\bar{v} = \bar{v}_c = 0$) and Σ_m (surface of the mobile parts, moving with them: $\bar{v} = \bar{v}_c \neq 0$). On Σ_w and Σ_m, $(\bar{v} - \bar{v}_c) \cdot \bar{n} = 0$.
- $\bar{n}$ is the outward normal of the control volume; $\bar{n}_{\text{in}} = -\bar{n}$ points into the fluid.

Definitions written on the surfaces, as in class:

$$\boxed{\bar{F} = \int_{\Sigma_m, \Sigma_w} (p - p_a)\, \bar{n}\, d\sigma - \int_{\Sigma_m, \Sigma_w} \bar{\bar{\tau}}' \cdot \bar{n}\, d\sigma = -F\, \bar{e}_x}$$

$$\boxed{\dot{Q} = \int_{\Sigma_w, \Sigma_m} \bar{q} \cdot \bar{n}_{\text{in}}\, d\sigma + \int_{V_c} Q_R\, dV}$$

$$\boxed{\dot{W} = -\int_{\Sigma_m} p\, \bar{n}_{\text{in}} \cdot \bar{v}\, d\sigma + \int_{\Sigma_m} (\bar{\bar{\tau}}' \cdot \bar{n}_{\text{in}}) \cdot \bar{v}\, d\sigma}$$

Only Σ_m enters $\dot{W}$: Σ_w is a fixed surface and does no work. $\dot{W}$ is the power of the gas on the moving solid, so the power done on the gas there is $-\dot{W}$.

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.10)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\,dV}}_{0\text{ (on average)}}+\int_{\Sigma_c}\rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma=0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int\rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma$
Σ_1	$-\rho_1 v_1 A_1$
Σ_2	$+\rho_2 v_2 A_2$
Σ_m	$0\ (\bar{v}=\bar{v}_c)$
Σ_w	0

$$\boxed{\rho_1 v_1 A_1=\rho_2 v_2 A_2=\dot{m}} \tag{1}$$

Linear momentum (Notes Eq. 3.35)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}}_{0}+\int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma=-\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma+\underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int\rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma$	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_1	$\rho_1 v_1\bar{e}_x(-v_1)A_1=-\dot{m}v_1\bar{e}_x$	$(p_1-p_a)A_1\bar{e}_x$	0
Σ_2	$\rho_2 v_2\bar{e}_x(v_2)A_2=\dot{m}v_2\bar{e}_x$	$-(p_2-p_a)A_2\bar{e}_x$	0

Surface	$\int \rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma$	$-\int (p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
$\Sigma_m,\ \Sigma_w$	0	$-\int_{\Sigma_m,\Sigma_w}(p-p_a)\,\bar{n}\,d\sigma$	$\int_{\Sigma_m,\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_m, Σ_w terms: $-\bar{F} = F\bar{e}_x$ (uniform velocity vectors give no viscous stress on Σ_1, Σ_2).

$$\boxed{\dot{m}\,(v_2 - v_1) = (p_1 - p_a)A_1 - (p_2 - p_a)A_2 + F} \tag{2}$$

Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\left(e+\frac{v^2}{2}\right)dV}_{0} + \int_{\Sigma_c}\rho\left(e+\frac{v^2}{2}\right)[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c}p\,\bar{v}\cdot\bar{n}\,d\sigma + \int_{\Sigma_c}(\bar{\bar{\tau}}'\cdot\bar{n})\cdot\bar{v}\,d\sigma + \underbrace{\int_{V_c}\rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect}} - \int_{\Sigma_c}\bar{q}\cdot\bar{n}\,d\sigma + \int_{V_c}Q_R\,dV$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e+\frac{v^2}{2}\right)[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int(\bar{\bar{\tau}}'\cdot\bar{n})\cdot\bar{v}\,d\sigma$	$-\int\bar{q}\cdot\bar{n}\,d\sigma$
Σ_1	$-\dot{m}\left(e_1+\frac{v_1^2}{2}\right)$	$+p_1v_1A_1$	0	$0\ (\nabla T=0)$
Σ_2	$+\dot{m}\left(e_2+\frac{v_2^2}{2}\right)$	$-p_2v_2A_2$	0	$0\ (\nabla T=0)$
Σ_m	0	$-\int_{\Sigma_m}p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int_{\Sigma_m}(\bar{\bar{\tau}}'\cdot\bar{n})\cdot\bar{v}\,d\sigma$	$-\int_{\Sigma_m}\bar{q}\cdot\bar{n}\,d\sigma$
Σ_w	0	$0\ (\bar{v}=0)$	$0\ (\bar{v}=0)$	$-\int_{\Sigma_w}\bar{q}\cdot\bar{n}\,d\sigma$

Boxed terms: the pressure and viscous work on Σ_m add up to $-\dot{W}$; the conduction through Σ_m, Σ_w plus the volume term $\int_{V_c} Q_R\,dV$ add up to $\dot{Q}$. Moving the pressure work on Σ_1, Σ_2 to the left gives $h = e + p/\rho$; with (1) and $h = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$:

$$\boxed{\dot{m}\left[\left(\frac{\gamma}{\gamma-1}\frac{p_2}{\rho_2}+\frac{v_2^2}{2}\right)-\left(\frac{\gamma}{\gamma-1}\frac{p_1}{\rho_1}+\frac{v_1^2}{2}\right)\right] = -\dot{W}+\dot{Q}} \tag{3}$$

(1), (2) and (3) determine ρ_2, p_2 and v_2.

PHASE 4

Interpretation, Limits and Dimensional Check

- (2) is the thrust balance: thrust = change of momentum flux + pressure imbalance on the sections. For $p_1 = p_2 = p_a$, $F = \dot{m}(v_2 - v_1)$.
- (3): the stagnation enthalpy of the stream rises by the heat added and falls by the work extracted, per unit mass flow.
- Dimensions (SI): $[\dot{m}(v_2 - v_1)] = \mathrm{N}$; $[(\dot{Q} - \dot{W})/\dot{m}] = \mathrm{J/kg} = \mathrm{m^2/s^2} = [h] = [v^2]$.
- Numerical illustration (hypothetical data): $A_1 = 0.5\ \mathrm{m^2}$, $A_2 = 0.3\ \mathrm{m^2}$, $v_1 = 150$ m/s, $p_1 = 0.9 \times 10^5$ Pa, $\rho_1 = 1.1\ \mathrm{kg/m^3}$ ($M_1 = 0.443$), $p_a = 10^5$ Pa, $F = 30$ kN, $\dot{W} = 2$ MW, $\dot{Q} = 30$ MW, $\gamma = 1.4$: $\dot{m} = 82.5$ kg/s and the two roots of the extension below, $v_2 = 355.515$ m/s, $p_2 = 126817$ Pa, $\rho_2 = 0.77353\ \mathrm{kg/m^3}$ ($M_2 = 0.742$), and $v_2 = 597.263$ m/s, $p_2 = 60336$ Pa, $\rho_2 = 0.46043\ \mathrm{kg/m^3}$ ($M_2 = 1.39$); residuals of (1) to (3) below 10^{-6} in SI units.
- Agreement with the official and class solutions: the same equations (1), (2), (3) (CL5.pdf writes (2) as $F = \rho_1 v_1 A_1 (v_2 - v_1) - (p_1 - p_a)A_1 + (p_2 - p_a)A_2$).

Extension (beyond the official solution): explicit solution

With $K = p_a A_2 + F + (p_1 - p_a)A_1 + \dot{m} v_1$, (2) gives $p_2 A_2 = K - \dot{m} v_2$ and, with (1), $p_2/\rho_2 = (K - \dot{m} v_2) v_2 / \dot{m}$. Substituting into (3) and using $\frac{1}{2} - \frac{\gamma}{\gamma - 1} = -\frac{\gamma + 1}{2(\gamma - 1)}$: $(\gamma + 1) v_2^2 - 2\gamma \frac{K}{\dot{m}} v_2 + 2(\gamma - 1) H_0 = 0$, with $H_0 = \frac{\gamma}{\gamma - 1}\frac{p_1}{\rho_1} + \frac{v_1^2}{2} + \frac{\dot{Q} - \dot{W}}{\dot{m}}$. The two roots are a subsonic and a supersonic exit state; for a subsonic entrance with heat addition the physical one is the subsonic (smaller) root, since reaching the other would require passing through sonic conditions inside the engine, which this overall balance cannot describe. For $F = 0$, $\dot{Q} = \dot{W} = 0$ and $A_1 = A_2$ one root is $v_2 = v_1$ and the product of the roots is $2(\gamma - 1)H_0/(\gamma + 1)$, the Prandtl relation of a normal shock.